A Novel Postprocessing Method for Robust Myoelectric Pattern-Recognition Control Through Movement Pattern Transition Detection

Bin Yu, Xu Zhang, *Member, IEEE*, Le Wu, Xiang Chen, *Member, IEEE*, and Xun Chen, *Senior Member, IEEE*

***Abstract*—Pattern-recognition-based myoelectric control systems are not yet widely available due to their limited robustness in real-life situations. Some postprocessing methods were introduced to improve the robustness in previous studies, but there is lack of investigation into movement transition phases. This article presents a novel postprocessing method based on movement pattern transition (MPT) detection. An image-based index is used to quantify the similarity of adjacent feature matrices from high-density surface electromyogram (EMG) signals. MPT detection is implemented by applying a double threshold to the calculated index. The proposed postprocessing method is used to rectify the EMG pattern recognition decisions from the classifier by incorporating the detected information. Two representative testing schemes are used to verify the robustness of the proposed method against force level variation and consecutive nonstop task performance. The proposed method achieved mean classification accuracy improvements of 7.33% and 10.91% with respect to the baseline performance of a raw classifier (without any postprocessing) in the two testing schemes. It also outperformed other common postprocessing methods ($p < 0.05$). Considering both the accuracy improvement and time efficiency for rapid responses to MPT, the proposed method could be a potential option for postprocessing to enhance the robustness of myoelectric control.**

***Index Terms*—Human–computer interface, movement pattern transition (MPT), myoelectric control, pattern recognition, postprocessing.**

Nomenclature

EMG	Electromyogram.
sEMG	Surface electromyography.
MV	Majority voting.
BF	Bayesian fusion.
LDA	Linear discriminant analysis.
MPT	Movement pattern transition.
HD-sEMG	High-density surface electromyography.
MVC	Maximal voluntary contraction.
PDI	Pattern distance index.
LDC	Linear discriminate classifier.
PSNR	Peak signal-to-noise ratio.
NMPT	Nontransition of movement pattern.
GO	Global optimization.
IO	Individual optimization.
MDMV	MPT-detection-based MV.
MDBF	MPT-detection-based BF.
ROC	Receiver operating characteristic.
TPR	True positive rate.
FPR	False positive rate.
TP	True positive.
FP	False positive.
TN	True negative.
FN	False negative.
AUC	Area under curve.
EER	Equal error rate.
RJRM	LDA classifier with a rejection option to remain in the latest decision.
ANOVA	Analysis of variance.

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The authors are with the Department of Electronic Science and Technology, University of Science and Technology of China, Hefei 230027, China (e-mail: yubinn@mail.ustc.edu.cn; xuzhang90@ustc.edu.cn; wule1994@mail.ustc.edu.cn; xch@ustc.edu.cn; xunchen@ustc.edu.cn).

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I. Introduction

Electromyogram (EMG) signals derived from muscle activities deliver important neural control information from the central nervous system. Myoelectrical signals are collected by electrodes placed on the surface of the skin in a noninvasive manner, which is called surface electromyography (sEMG). Myoelectric control has been used extensively to establish an alternative interface for voluntarily manipulating external devices, including power exoskeletons/orthoses [1], [2], rehabilitation robots [3], and power prostheses [4]. Myoelectric pattern recognition has been successfully investigated in the control of multiple degrees of freedom for multifunctional applications [3]–[6]. High classification accuracies have been reported [3]–[11], but they were primarily obtained under ideal laboratory conditions.

Robust performance in real-world settings is a prerequisite to the wide application of the myoelectric control [12]. Due to the nonstationary characteristics of EMG, the continuous signal stream needs to be segmented into a series of analysis windows with sufficiently long duration to characterize the EMG pattern for classification. A longer analysis window appears to

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be beneficial for improving the myoelectric pattern recognition performance and robustness [13], [14], but it incurs a greater time delay [15]. This trade-off has to be considered when implementing algorithms for real-time myoelectric control [13], [14], [16].

Besides the transient nature of EMG, several disturbance factors contribute to the differences between laboratory and real-world conditions, which may also compromise the robustness of a myoelectric pattern recognition system. These factors include electrode shift [17]–[21], force variation [22], and limb orientation [23], [24]. Many studies have investigated their effects to alleviate the performance degradation that they cause.

Recently, many efforts have been made toward the robustness and usability of myoelectric pattern recognition control systems. Postprocessing methods play a very important role in this context. Postprocessing comprises techniques that are applied after the main myoelectric pattern recognition to correct or decrease the negative effects of misclassifications using certain strategies. Postprocessing has proven to be very helpful in enhancing robustness when applied in real-time control systems.

Postprocessing methods can be roughly divided into three main categories according to their strategies. The first category uses a multiwindow joint decision strategy to rectify decision about the current window based on a sequence of decisions about multiple consecutive windows. Both the majority voting (MV) [6], [15] and Bayesian fusion (BF) [25], [26] methods are in this category. As the most conventional example, Englehart and Hudgins [15] introduced the MV method to correct some misclassifications. This method is a viable option to enhance controllability, especially when considering its simple implementation. The second category uses a confidence score to determine whether the current decision from the classifier is reliable [27], [28]. For example, Scheme *et al.* [27] proposed an extended linear discriminant analysis (LDA) classifier that is capable of rejecting unreliable decisions, but the classification accuracy was not improved very much. By combining both categories, Amsuss *et al.* [29] proposed a postprocessing method using an artificial neural network to fit the confidence score to the classification results of multiple windows. However, this method contains numerous parameters that need to be trained. Methods in the third category are designed to tolerate some misclassifications directly and employ a special control strategy like the velocity ramp method [30] to decrease their effect on the control variable.

The state-of-the-art postprocessing methods improve the performance of myoelectric pattern recognition by enhancing the stability of the classifier's output and eliminating jitter (possible errors) within the decision stream over a long duration. Overcoming most interference is feasible when myoelectric patterns are implemented in a continuous and relatively slow manner for control. Due to such stability, however, the postprocessing approach is likely to fail to respond rapidly to new movement patterns, resulting in fault decisions or inevitable time delays. This can explain an existing issue of classification errors mostly being concentrated around movement pattern transitions (MPTs) when the postprocessing is performed [29]. Therefore, the true MPT information needs to be considered and involved in the design of a postprocessing method for myoelectric pattern recognition control.

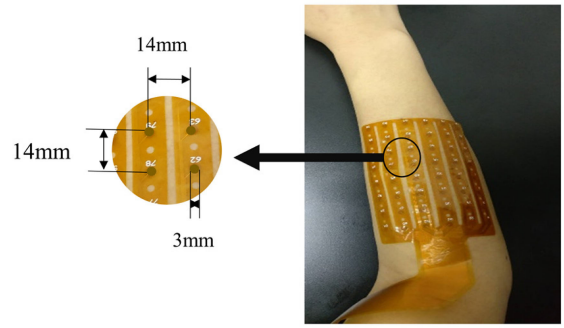

Fig. 1. Experimental setup for recording sEMG data with a high-density electrode array.

In this article, a novel postprocessing method based on MPT detection is proposed. Incorporating prior detected information can improve the robustness of myoelectric pattern recognition control in terms of both accuracy and responsiveness, especially when a true MPT takes place. In the proposed method, MPT is detected using a novel image-based index that represents how the movement patterns are changed. This is done by taking advantage of spatial information from high-density sEMG (HD-sEMG), which better characterizes the movement patterns by means of muscle imaging [3], [20], [21]. Two kinds of interference were examined in testing datasets to demonstrate the effectiveness of the proposed method: force level variation and nonstop consecutive task performance (no rest between two performed tasks). The proposed method could offer a practical solution for enhancing the robustness of myoelectric pattern recognition control.

II. Methods

A. Subjects

Nine able-bodied subjects with intact limbs were recruited (seven males, two females, age 23.7 $\pm$ 1.3 years). This study was approved by the local ethics committees, and each subject signed an informed and written consent form.

B. Experiments

A flexible 6 $\times$ 8 array of electrodes was placed around the forearm of the subjects approximately 5 cm distal to the olecranon of the elbow (see Fig. 1). Each electrode probe had a diameter of 3 mm, and the inter-electrode distance was 14 mm. Each electrode was operated in a monopolar configuration for one-channel EMG recording. The HD-sEMG data were recorded by a custom-made data recording system. The raw data were amplified with a total gain of 60 dB, bandpass filtered at a width of 20–450 Hz, and sampled at 1 kHz by 16-b digitalization. The recorded data were transferred to a computer via a USB cable, and software with a graphical user interface was developed to monitor and process all the recorded EMG data on the computer screen in real time.

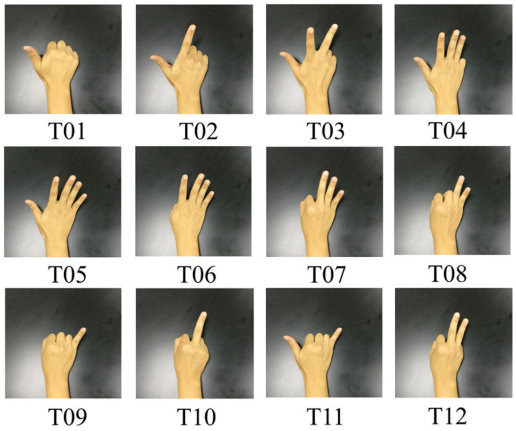

Fig. 2. Images of the 12 movement tasks.

During the experiment, the subjects sat on a chair with their forearms comfortably placed on a height-adjustable side table. We examined 12 gestural tasks of the wrist and hand involving the extension of different individual fingers or finger combinations, as shown in Fig. 2. When considering relaxation/rest as one "task" as well, there were 13 tasks in total for classification. The entire procedure consisted of two experimental scenarios.

1) MPTs With Relaxation: In the first scenario, each task implementation started from and ended in a relaxation state, which is commonly used for conventional myoelectric control studies. This experimental scenario consisted of 12 trials. In each one, the subjects were instructed to perform one of the 12 tasks with eight repetitions at two different force levels. The first four repetitions were at a relatively low force level (denoted as *L*), which corresponded to mild muscle contraction strength at about 30–40% maximal voluntary contraction (MVC). The later four repetitions were at a higher force level (denoted as *H*), which corresponded to strong muscle contraction strength at about 60–70% MVC. After completing each task, the hands and fingers slowly returned to a neutral position and remained there with all fingers naturally bent. A video clip was prepared in advance and played as a guideline to instruct the subjects in how to implement tasks and their timings. Following the video guidance, the subjects were asked to hold each repetition of the task for 6 s with a relaxation period of 4 s between two consecutive repetitions.

2) MPTs Without Relaxation: The second scenario required continuous performance of a series of tasks without relaxation between two consecutive tasks. Moreover, all data collected in this scenario were processed in real time. More details about the classification program are described in the following section.

Five tasks were employed (T01–T05). The subjects were instructed to perform the five tasks at a mild, comfortable force level in the order of their task index number through successional extension of additional one finger from the thumb to the little finger. This series of task performances was used to test the dexterous control of the fingers representing the basic functions involved during typing on a keyboard. Similarly, video guidance was used to instruct the subjects to hold each task for 5 s and then to transition to the next one without relaxation. Four trials were conducted so as to repeat the procedure for four times.

During both experimental scenarios, the true MPTs needed to be marked down on the timeline. Although some routine computerized methods could help (e.g., EMG onset/offset detection using amplitude thresholding [3], [25], [31]), visual inspection by an experimenter was performed through all the recorded data. This MPT information was used as the ground-truth for testing the MPT detection performance.

C. Data Analysis

Fig. 3 shows a flowchart of the proposed postprocessing method. The method calculates a pattern distance index (PDI) to detect MPTs. Based on the obtained MPT information, a novel postprocessing framework was established to rectify and update the decision sequence from the classifier. The analysis was performed on a workstation with a 3.41 GHz Intel Core i5 CPU in MATLAB (version R2016b, MathWorks, Natick, MA, USA).

1) Feature Extraction: A variety of features have been used to characterize raw EMG data for pattern classification. In particular, four time-domain features proposed by Hudgins *et al.* [8] have been frequently used as a time-domain feature set due to their effectiveness and low computational cost: the mean absolute value, zero crossing, slope sign change, and waveform length [7], [8], [27], [32], [33]. Therefore, four features per channel were calculated from each interval (i.e., analysis window) of 128 ms with increments of 64 ms (with 64-ms overlap). Subsequently, a $6 \times 8 \times 4$ feature matrix was formed over the entire electrode array for each analysis window. Each feature matrix could also be reshaped into a 192-dimensional (192-D) feature vector that is suitable for input into the following classifier.

2) Classification: A conventional linear discriminate classifier (LDC) was used for task classification due to its satisfactory performance and high practicability that is comparable with more complex classification methods [34]–[36]. The LDC used in this study involved two steps. The first step is LDA to minimize the within-class distance and maximize the between-class distance through an optimal linear transformation. Specifically, a multivariate extension of LDA called uncorrected LDA [8] was used due to the high dimension of feature vectors derived from HD-sEMG recordings. The second step is to construct a linear classifier by modeling the within-class density of each class as a multivariant Gaussian distribution [36], [37].

3) Proposed Postprocessing Method:

Stage 1: MPT detection

A PDI is proposed to account for the similarity of adjacent windows

$$\mathrm{PDI}\left(x\right) = psnr^{3}\left(S\left(x-1\right), S\left(x\right)\right) \tag{1}$$

where S is the 3-D feature matrix of an analysis window, and x is the analysis window index. Here, *psnr* means to calculate the peak signal-to-noise ratio (PSNR) between two matrices. In image quality evaluation, the PSNR is the ratio between the

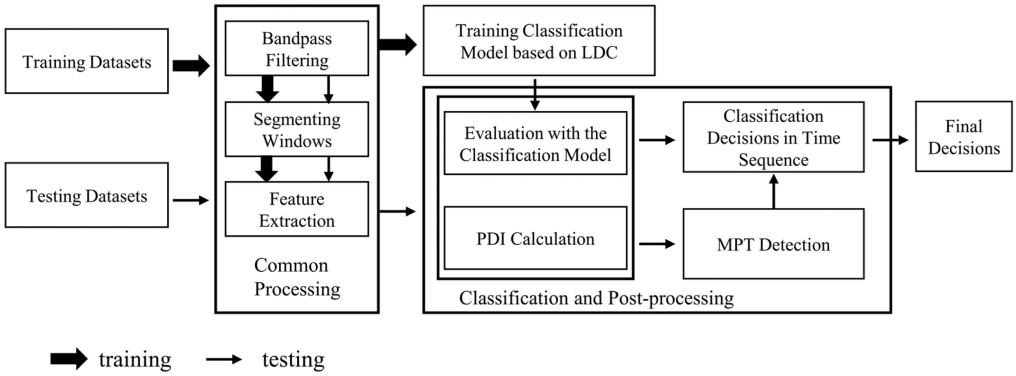

Fig. 3. Flowchart of the proposed method.

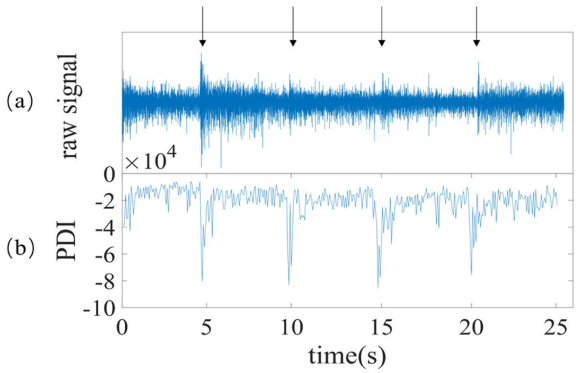

Fig. 4. Example of PDI varying with the raw signals. (a) Raw signals collected by a randomly selected channel during one trial in the scenario for MPTs without relaxation. The four black arrows roughly indicate the moments of MPT. (b) PDI calculated from signals.

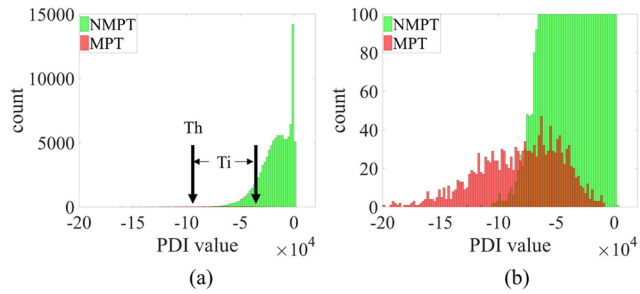

Fig. 5. Histogram of the PDI value, grouped by MPT and NMPT. (a) Overall PDI value distribution. Since the number of occurrences of MPT is much less than that of NMPT, the vertical axis is enlarged in (b).

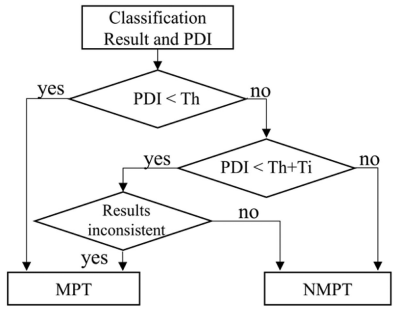

Fig. 6. Schematic diagram of the MPT detection procedure.

maximum possible power of an image and the power of corrupting noise that affects the fidelity of the image representation.

In this study, the change of the muscle contraction pattern as a result of task transition is analogous to a change of the image pattern characterized by HD-sEMG. A transition to a new pattern is assumed to impose some inevitable noise on the previous pattern. This is the reason for choosing the PSNR to reflect MPTs. Its third power was further calculated as the PDI after some pretests because it expanded the difference of the distribution between MPT and nontransition of movement pattern (NMPT).

PDI exhibited a valley at the moments of MPT with very high sensitivity, but it was not as sensitive to other types of interference (e.g., muscle force variation). Fig. 4 shows an example of the PDI varying with the raw signals during one trial in the scenario for MPTs without relaxation. The PDI curve dropped during the transition phases. The distribution of PDI is greatly different between MPT (-8.14 ± 3.61E4) and NMPT (-1.64 ± 1.42E4) with data collected in both scenarios across nine subjects (see Fig. 5). Therefore, a double-threshold method was designed to differentiate between MPT and NMPT.

To appropriately set both thresholds, we defined a basic threshold Th and a threshold increment Ti. Th is used to definitively determine MPT when a lower PDI takes place. Ti describes the overlapping range of PDI for both MPT and NMPT. If a PDI is greater than Th+Ti, the tested window is considered to have the same pattern as previous ones. For PDIs between Th and Th+Ti, the determination of MPT needs to rely on the classifier's decision as well. As shown in Fig. 6, MPT is detected for an analysis window with PDI between Th and Th+Ti if its classification decision is inconsistent with those of previous ones.

The setting of both thresholds has a great influence on the MPT detection performance. Sensitivity analyses were performed in pretests for both thresholds by adjusting Ti within a range of 0 to 7E4 and Th from -1.40E5 to -0.30E5. Both thresholds were optimally determined according to the MPT detection performance and task classification. We also used two strategies for determining Th to examine its versatility.

1) Global Optimization (GO): One Th was selected for all tested subjects to allow optimal overall performance.
2) Individual Optimization (IO): Each Th was selected to be specific to each subject to allow performance optimization for individual subjects.

Stage 2: Decision Rectification

Given the detected MPT information, the decision from the classifier is rectified to be a final decision for the current window in this stage. Therefore, we established a decision queue with a variable length from 0 to n, to store decisions of the classifier for a set of previous windows. If an MPT is detected, the decision queue is emptied; the classifier decision is regarded as the final one. Otherwise, the classifier decision, along with the queue of previous decisions, is processed through common methods for multiwindow joint decision making to produce a final decision. There are two options: the MV and BF methods. The classifier decision is then put into the queue. The oldest decision is removed if the queue is full.

The MV approach employs n previous classification results with the current one d_i. The value of d_{mv} is used to simply choose the class with the greatest number of occurrences in the $n+1$ analysis windows of the decision stream. The process rejects misclassifications and ensures smooth operation. In a preliminary study, when n was equal to 8, the highest recognition accuracy was achieved based on satisfying the response delay requirement (250 ms in the case of nine votes [15]).

The BF approach was suggested as an alternative to MV that uses a disjoint windowing scheme for feature extraction [25]. We extended the use of BF to the overlapping windowing scheme, which has been demonstrated to be effective [26] due to the nonstationarity of the EMG signals. The BF approach uses the posterior probabilities generated by the classifier. Feeding the features extracted from each window to the classifier results in M conditional probabilities $p(C_i|W_n)$, where $i = 1, 2, 3, \ldots, M$, which correspond to the probability that the analysis window W_n belongs to a certain class C_i in a problem with M classes. Instead of simply counting the number of occurrences of a certain class label, the BF operates by multiplying the resulting probabilities

$$p\left(C_i|W_1, W_2, \ldots, W_{n+1}\right) = \triangle \prod_{k=1}^{n+1} \left(p\left(C_i|W_k\right) + T_j\right) \quad (2)$$

where $\triangle$ is a normalized constant to make $p(C_i|W_1, W_2, \ldots, W_{n+1})$ a probability density function, T_j is a bias parameter to avoid multiplying by zero, and j is the location of the current window in a queue of the previous n windows. T_j is defined as [25]

$$T_j = 10 \times \frac{\exp\left(-0.5 \times j/\left(n+1\right)\right)}{\sum_{k=1}^{n+1} \exp\left(-0.5 \times k/\left(n+1\right)\right)} \quad (3)$$

where $j = 1, 2, 3, \ldots, n+1$, with the last window inserted in the queue taking the first position of 1.

In the proposed framework, both the MV and BF algorithms can be used for the multiwindow joint decision, thus producing an MPT-detection-based MV (MDMV) algorithm and an MPT-detection-based BF (MDBF) algorithm, respectively.

TABLE I
ILLUSTRATION OF TERMS FOR MPT DETECTION

Predicted outcome	Condition (as determined by the standard of truth)	
	Positive (MPT)	Negative (NMPT)
Positive	True positive (TP)	False positive (FP)
Negative	False negative (FN)	True negative (TN)

D. Performance Evaluation and Statistical Analysis

To demonstrate the effectiveness of the postprocessing method, two testing schemes involving the two kinds of interferences were designed. One is a cross-level testing scheme to introduce force level variation. In the scenario for MPTs with relaxation, the data recorded from each trial were divided according to the force level (L or H). This cross-level testing scheme was conducted as L–H and H–L, where both sides of the en-dash (–) indicate the levels for training and testing, respectively. The other was a nonstop testing scheme, where all data recorded in the scenario for MPTs without relaxation were used for testing, given a well-trained classifier established with the training dataset consisting of all data from the scenario for MPTs with relaxation. Thus, possible data variation during nonstop consecutive task performance could be regarded as interference in the task classification, while the classifier was trained using data with the presence of relaxation/rest between gestural task performances. This testing scheme was implemented online, as described in Section II-B.

The MPT detection is a prerequisite for the proposed postprocessing method. To evaluate its performance and to establish criteria for setting the necessary thresholds, a receiver operating characteristics (ROC) curve was obtained, where the true positive rate (TPR) was plotted against the false positive rate (FPR) at various thresholds. Their definitions are based on four terms: true positive (TP), false positive (FP), true negative (TN), false negative (FN), as explained in Table I. On this basis, the following metrics are defined:

$$\mathrm{TPR} = \frac{\#\mathrm{TP}}{\#\mathrm{TP} + \#\mathrm{FN}} \quad (4)$$

$$\mathrm{FPR} = \frac{\#\mathrm{FP}}{\#\mathrm{FP} + \#\mathrm{TN}} \quad (5)$$

$$\mathrm{Accuracy} = \frac{\#\mathrm{TP} + \#\mathrm{TN}}{\#\mathrm{TP} + \#\mathrm{FP} + \#\mathrm{TN} + \#\mathrm{FN}}. \quad (6)$$

Here, the symbol # represents the number of incidents. There are two metrics on the opposite side of TPR and FPR: true negative rate (TNR) and false negative rate (FNR), respectively. The relation among these four metrics is as follows:

$$\mathrm{TPR} + \mathrm{FNR} = 1 \quad (7)$$

$$\mathrm{FPR} + \mathrm{TNR} = 1. \quad (8)$$

Despite using a rigorous video timing approach, there are inevitable deviations in the recorded true MPT information

because dynamic data are involved in multiple consecutive windows. Therefore, a three-window tolerance was employed when evaluating the MPT detection performance. The correctness of the MPT detection was based on a criterion where the analysis window decided with the occurrence of MPT has to be no farther than three windows from the window where the true MPT takes place for the first time. The MPT detection performance was quantified by two metrics of the ROC curve: the area under curve (AUC) and the equal error rate (EER). EER is the point where TPR is equal to FPR. Both lower EER and higher AUC indicated better performance of the MPT detection. Therefore, Ti was determined for the optimal performance with the training dataset. Besides, the accuracy presented in (6) is an alternative, single-number summary of the MPT detection performance.

Regarding the task classification performance, the identification of the "relaxation" pattern was always correct. Therefore, the relaxation patterns were not taken into account when evaluating the performance of the classifier to avoid biased high performance. Thus, the classification accuracy was calculated as

$$\mathrm{Acc} = \frac{\#\mathrm{Correct\ Decisions}}{\mathrm{Total}\ \#\mathrm{Decisions}} \times 100\%. \tag{9}$$

This is also called the total accuracy presented by Peerdeman *et al*. [38].

Four routine classification and postprocessing methods were employed for comparison. The baseline performance was defined using the raw LDC classifier without postprocessing. We also implemented conventional postprocessing methods that use the multiwindow joint decision strategy. They were MV and BF combined with the LDC. In addition, the RJRM method was also used as a representative approach that relies on the strategy to evaluate the reliability of the classifier's outputs. It linearizes the class likelihoods produced by the LDC and normalizes their sum to 1. If the maximum probability of a class is lower than 0.97 [27], the classifier's decision is rejected, and this unreliable decision is rectified as the latest accepted classification decision. All the methods selected for performance comparison were compatible and suitable for the data used in this study, and they were fine-tuned for a fair comparison.

To verify the effectiveness of the proposed postprocessing method, a one-way repeated-measure analysis of variance (ANOVA) was applied to the classification accuracy (averaged over both H–L and L–H tests), which was obtained in the cross-level testing scheme. The algorithms were considered as the within-subject factor (six levels: Baseline, MV, BF, RJRM, MDMV, and MDBF). A one-way repeated-measure ANOVA was also conducted on the classification accuracy obtained in the nonstop testing scheme to examine the performance of the six algorithms.

In addition, time delays of responding to MPTs were also computed with the data used in the cross-level testing scheme, and a one-way repeated-measure ANOVA was conducted to examine the performance of the six algorithms in terms of rapid response. The level of statistical significance was set to $p < 0.05$ for all analyses. Post hoc pairwise multiple comparisons with Bonferroni correction were used. All statistical analyses were completed using SPSS software (ver. 24.0, SPSS Inc., Chicago, IL, USA).

III. Results

A. MPT Detection Performance

The effect of Ti on the MPT detection performance was examined with data collected in the scenario for MPT with relaxation, as shown in Fig. 7. Fig. 7(a)–(d) shows four representative examples of the ROC curves for nine subjects when the Ti was 0, 2E4, 4E4, and 6E4. Increasing Ti led to improved performance. This trend was confirmed by the curves of AUC and EER plotted against Ti in Fig. 7(e) and (f). AUC increased and EER decreased when Ti was increased at first. Afterward, both curves remained constant when Ti was larger than 3.5E4, indicating that the MPT detection performance stably reached a high level (AUC: 0.9984, EER: 0.0114, and accuracy over 95%, accordingly). Therefore, the value of Ti was set to 4E4 for other analyses.

Fig. 8 shows the effect of Th on the classification accuracy averaged over nine subjects. The MDMV method under the cross-level testing scheme was used as a representative example. Unsurprisingly, when increasing Th from −1.40E5 to −3.0E4, the classification performance climbed at first, remained stable at a high level (about 84.15%), and finally dropped. When Th was selected within the range of −1.0E5 to −0.8E5, an optimally high level of the averaged classification accuracy could be retained (approximately 85%). Therefore, Th was set as −0.92E5 for the following analyses.

B. Classification Performance

1) Cross-Level Testing Scheme: Fig. 9 shows the classification accuracies averaged over all subjects for the six methods under the cross-level testing scheme. Obviously, when the force variation was presented in the testing datasets, the baseline performance of the myoelectric pattern recognition control achieved really low accuracies of 77.17 ± 8.87% in the L–H test and 76.90 ± 5.96% in the H–L test. Unsurprisingly, all other algorithms with postprocessing achieved significantly improved performance ($p < 0.01$) when both tests were considered together.

Within the proposed framework, both MDMV and MDBF achieved accuracies of 85.80 ± 7.77% and 86.61 ± 6.67% for both tests, with significant improvement over the baseline performance by 7.33% and 8.24% ($p < 0.001$). Each one significantly outperformed all the other common postprocessing methods ($p < 0.001$). Table II lists the classification accuracies achieved by both the MDMV and MDBF methods using the two strategies for determining Th. Both the GO and IO strategies had comparable classification accuracies for any method and any test under this testing scheme, with no significant difference ($p = 0.886$).

2) Nonstop Testing Scheme: The nonstop testing scheme was carried out with an online control system, and data from four trials for each subject were pooled together for the performance evaluation. Fig. 10 shows the classification accuracies averaged over all subjects for the six methods. All postprocessing algorithms performed significantly better than the baseline method

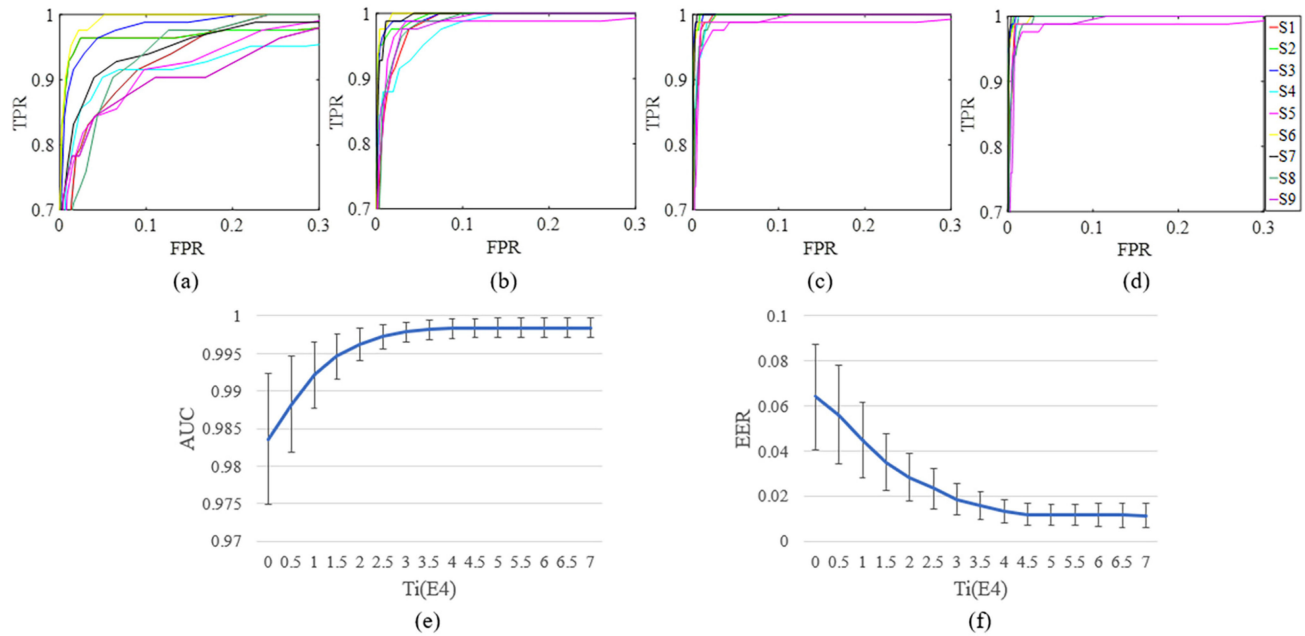

Fig. 7. Performance sensitivities to the threshold Ti. (a)–(d) Resultant ROC curves of MPT detection for nine subjects when the Ti was set to 0, 2E4, 4E4, and 6E4, respectively. (e) Mean AUC value as a function of Ti. (f) Mean EER value as a function of Ti. Error bars represent standard deviations.

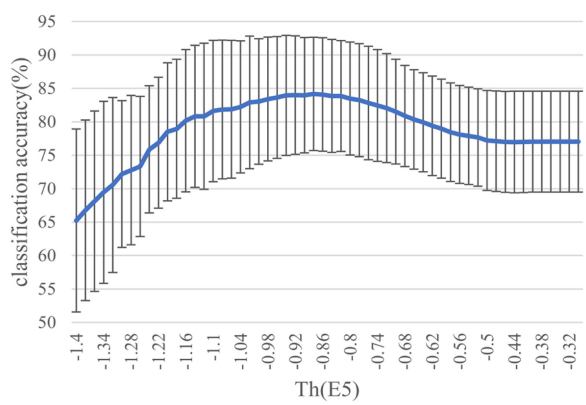

Fig. 8. Classification accuracy of MDMV varies with Th when Ti is set to 4E4 in the scenario for MPTs with relaxation. The accuracies are averaged across nine subjects and plotted with standard deviations error bars.

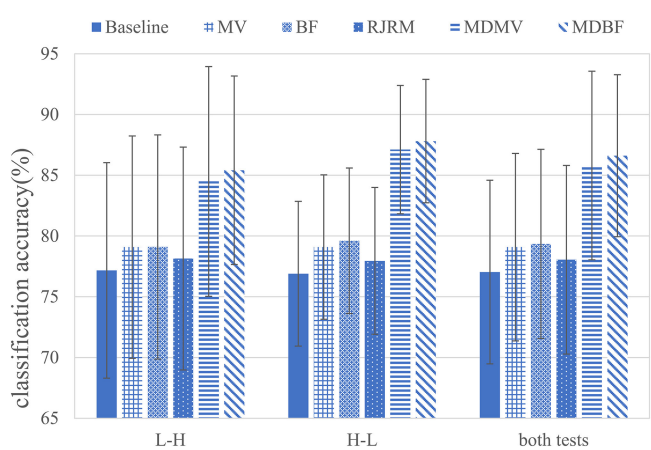

Fig. 9. Classification accuracies averaged over nine subjects for L–H, H–L and both tests in the cross–level test. Error bars represent standard deviations.

TABLE II
DETAILS OF DIFFERENCE ON CLASSIFICATION ACCURACY BETWEEN GO AND IO

		GO	IO
MDMV	L–H	84.49%	84.97%
	H–L	87.11%	88.09%
MDBF	L–H	85.41%	86.23%
	H–L	87.81%	88.86%

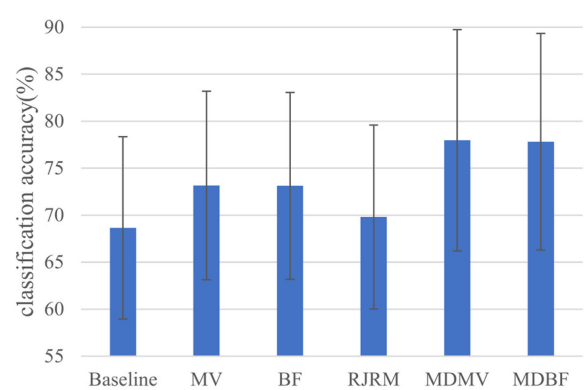

Fig. 10. Classification accuracies averaged over nine subjects in the scenario for MPTs without relaxation using six methods, respectively. Error bars represent standard deviations.

($p < 0.05$). Both MDMV (77.97 ± 11.76%) and MDBF (77.81 ± 11.52%) outperformed all other methods ($p < 0.05$). Multiple pairwise comparisons revealed that each of them achieved significantly better performance than all other methods ($p < 0.05$).

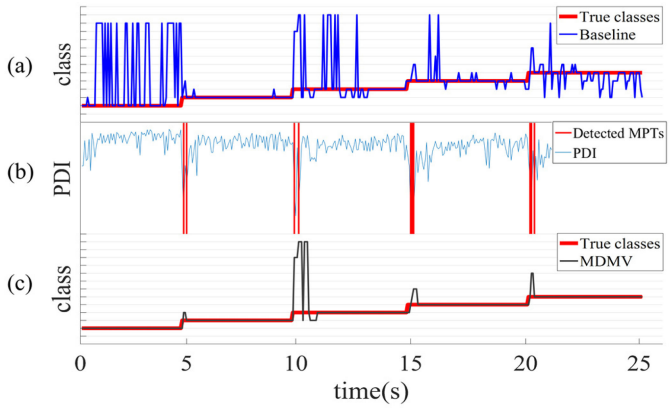

Fig. 11. Example of rectifying the misclassifications of LDC using MDMV in the scenario for MPTs without relaxation. (a) Decision sequence from the raw LDC for classification of 13 tasks including relaxation. (b) PDI (blue line) calculated from the corresponding raw signals and the detected MPTs marked by vertical red lines. (c) Corresponding sequence of decisions rectified by the proposed MDMV method.

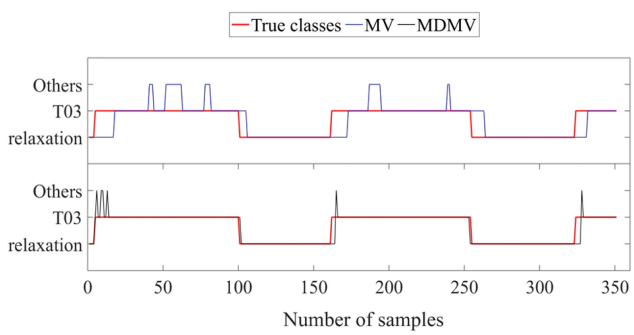

Fig. 12. Example of decision sequences for identifying the task T03 using both MV and MDMV methods in the scenario for MPTs with relaxation. The sequence of true classes is also plotted. All other tasks except T03 and relaxation are referred to as "Others."

Fig. 11 shows a representative example of rectifying the classifier's decision using the MDMV method under the nonstop testing scheme. Consecutive implementation of tasks without any relaxation led to frequent errors produced by the raw classifier without postprocessing with respect to the true classes. Given the correct detection of all MPTs, the proposed MDMV method was able to correct most errors from the classifier, thus yielding a decision sequence that approximates the sequence of true classes.

C. Time Delay

The response time delay with respect to the true task sequence was examined as well. Fig. 12 illustrates the decision sequences for identifying an example task (T03) in the scenario for MPTs with relaxation using both the MV and MDMV methods. Obviously, the MV always needed to process more samples or windows to respond to every true MPT. On this basis, a time delay can be quantified by the number of windows that it took to respond to each MPT.

Fig. 13 shows the time delays of the six methods. Both the baseline and the RJRM methods were able to yield the lowest

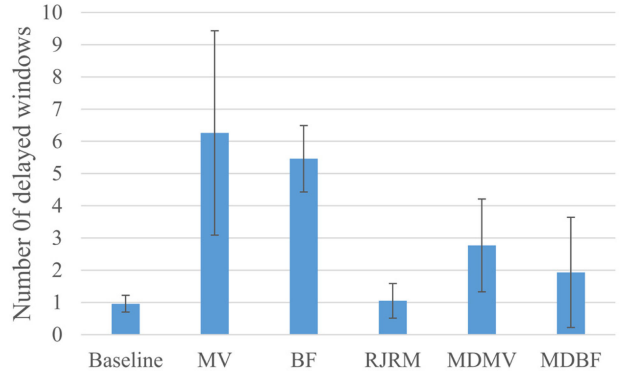

Fig. 13. Time delays quantified by the number of windows that it took to respond to new movement patterns, averaged over nine subjects in the cross-level testing scheme, using six methods, respectively. Here, one window represents a time delay of 64 ms (equivalent to a window increment). Error bars represent standard deviations.

time delays of around one window (=64 ms, $p < 0.001$). Both the conventional MV and BF methods had the greatest time delays over five windows (320 ms) among all six methods ($p < 0.001$). In contrast, the proposed MDMV and MDBF achieved a significant reduction in time delay to less than three windows (192 ms, $p < 0.001$). Such a late response to MPTs over multiple consecutive windows was previously counted as a series of classification errors already. Regarding the computational time cost, all methods were able to produce a final decision for each window within 20 ms. This computational time delay, along with the window length of 128 ms, was still no longer than the noticeable limit of 175 ms, as indicated by Farrell and Weir [16].

IV. Discussion

We have proposed a novel postprocessing method for improving the robustness of myoelectric pattern recognition control by incorporating the information of MPTs. Each MPT is accompanied by a muscular activation change. Such alterations could be directly reflected by the proposed image-based PDI, as confirmed by Figs. 4 and 5. Therefore, the classification performance and robustness were improved, especially when the MPT takes place. The MPT detection method can be regarded as an extension of EMG onset/offset detection, which aims to differentiate EMG activities from the baseline at rest. This can be easily confirmed in the experiment for MPT with relaxation, where each detected MPT was equivalent to the corresponding onset or offset time of task performance.

The routine onset/offset detection has always been performed by applying single or double thresholds to the EMG amplitude [3], [25], [31], and it is likely to be affected by various interferences. In the proposed method, the spatial information derived from HD-sEMG was used to replace the amplitude of an individual channel, allowing better characterization of the muscular activation pattern transitions. Therefore, the proposed method not only yields robust performance against possible interferences, but also has the potential to be applied to EMG interpretations during nonstop task performances, such as piano playing, keyboard typing, and communicating with sign

language [15]. Both advantages of the proposed method were demonstrated in this study.

Through performance sensitivity analyses, both thresholds involved in the MPT detection could be appropriately selected to ensure accuracies over 95% for discrimination between MPT and NMPT (not task classification accuracies). This sets up a good basis for improving the classification performance with the proposed postprocessing method. Furthermore, both IO and GO strategies for selecting thresholds individually and globally yielded almost the same thresholds and comparable performance in terms of classification accuracy, with no significant difference (see Table II). This indicates good versatility of the proposed MPT detection method. Therefore, it is straightforward to transfer the determined thresholds to a new user without any calibration.

When the force variation and absence of relaxation for continuous task performances were presented in the testing datasets, the raw classifier for myoelectric pattern recognition control achieved mean accuracies of $77.03 \pm 7.56\%$ and $68.64 \pm 9.7\%$, which were compromised in comparison to the performance under well-controlled laboratory conditions [3]–[11]. All postprocessing methods achieved significantly improved classification accuracy ($p < 0.05$) over the baseline. Both versions of the proposed method significantly outperformed other common postprocessing methods ($p < 0.001$).

The proposed method also has a rapid response to a new pattern in comparison with conventional methods (i.e., MV and BF) that rely on the multiwindow joint decision strategy. This is a result of the incorporation of MPT information. During the NMPT period, the conventional multiwindow joint decision approach was implemented to ensure relatively stable output decisions and to overcome external interferences (see Fig. 11). When a MPT took place and was correctly detected, the new pattern could be responded to instantaneously with a short time delay (see Fig. 12). This approach could avoid incorrect rectification of the current decision from the new pattern to a previous pattern due to a majority of previous decisions. With such a modification, the proposed postprocessing methods were proven to significantly outperform the conventional MV and BF methods in terms of shorter time delays at the moment of MPTs ($p < 0.001$). It is evident that the baseline method without any postprocessing and the RJRM method with no multiwindow joint decision strategy achieved the shortest time delay (see Fig. 13). The inability of the proposed method to respond to MPTs as fast as the baseline method can be attributed to the time delays incurred by the MPT detection (i.e., the difference between the true MPT and its detected timing). Considering the trade-off between classification accuracy and responsiveness of the system, the proposed method is demonstrated to improve the classification accuracy greatly at the expense of a little prolonged response.

Please also note that the MPT detection performance evaluation and the time delay computation referred to the prelabeled MPT information in the form of true class sequence. In fact, a certain amount of inevitable errors in these labeled MPTs were allowed due to difficulty in sensing exact timings of MPTs. This is also the reason for explaining why a three-window tolerance was employed when evaluating MPT detection performance. However, such information and the labeled class sequence were believed to approximate to the ground-truth within the scope of human control capability. Therefore, the performance comparison among different methods was still statistically meaningful.

It should be acknowledged that there are some limitations in the current study. The PDI for MPT detection was calculated using data from a high-density electrode array without a corresponding exploration of sEMG data from separate recording sites. Furthermore, only able-bodied and neurologically intact subjects were recruited. The next stage is to enhance the clinical relevance of the proposed method by applying it to amputees or subjects with stroke. All of these topics will be the focuses of our future work.

V. Conclusion

In this article, a novel postprocessing method was proposed for improving the performance of a myoelectric pattern recognition system. The results were compared to those of other methods. The method incorporating with MPT information is a modification of routine postprocessing methods that rely on multiwindow joint decision making. In the two representative testing schemes, the proposed method outperformed other common methods with a significant improvement in classification accuracy, especially during the transition phases. It also exhibited rapid response to true MPTs, which is superior to conventional methods with multiwindow joint decision making. Therefore, the method is shown to be a desirable enhancement for myoelectric pattern recognition control systems.

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